22. Ratio & Proportion

Year 4 · **Supplementary Worksheet** — extra practice & worked breakdowns

Worked Examples

Example 1 — Scaling a recipe:

A recipe needs 3 eggs for every 1 cake. How many eggs are needed for 5 cakes?

Answer: Multiply by 5: 3 × 5 = **15 eggs**.

Example 2 — Sharing in a given ratio:

Share 15 sweets in the ratio 2:3.

Answer: 2 + 3 = 5 parts. 15 ÷ 5 = 3 per part. Shares: 2 × 3 = 6 and 3 × 3 = **9**, so **6 and 9**.

Example 3 — Direct proportion (unit method):

A recipe for 5 people needs 250g of flour. How much flour is needed for 8 people?

Answer: Find the amount for 1 person: 250 ÷ 5 = 50g. Then scale up: 50 × 8 = **400g**.

Example 4 — Scale drawings:

A model car is built at a scale where 1 cm represents 12 cm in real life. If the model is 5 cm long, how long is the real car?

Answer: Multiply by the scale factor: 5 × 12 = **60 cm**.

Example 5 — Rate problems:

A tap fills a container at a steady rate. In 4 minutes it fills 10 litres. How many litres will it have filled after 6 minutes?

Answer: Find the rate per minute: 10 ÷ 4 = 2.5 litres/min. Then scale up: 2.5 × 6 = **15 litres**.

Foundation

15 questions

1. A recipe for 1 person needs 6 biscuits. How many biscuits for 3 people?

2. Scale up the ratio 2:5 by multiplying both parts by 3. What is the new ratio? A) 5:8 B) 6:15 C) 6:8 D) 2:15

3. A pattern uses 5 blue beads for every 1 red bead. If there are 2 red beads, how many blue beads are there?

4. Share 10 sweets equally between 2 people. Select ALL statements that are TRUE: A) Each person gets 5 sweets B) Each person gets 4 sweets C) This is sharing in the ratio 1:1 D) This is sharing in the ratio 2:1 E) Together they get 10 sweets

5. A recipe needs 2 eggs for every 1 cake. How many eggs are needed for 5 cakes?

6. A recipe for 1 person needs 5 biscuits. How many biscuits for 4 people?

7. Scale up the ratio 3:5 by multiplying both parts by 3. What is the new ratio? A) 6:10 B) 9:15 C) 9:10 D) 3:15

8. A pattern uses 2 blue beads for every 1 red bead. If there are 5 red beads, how many blue beads are there?

9. Share 12 sweets equally between 2 people. Select ALL statements that are TRUE: A) Each person gets 6 sweets B) Each person gets 5 sweets C) This is sharing in the ratio 1:1 D) This is sharing in the ratio 1:2 E) Together they get 12 sweets

10. A recipe needs 4 eggs for every 1 cake. How many eggs are needed for 3 cakes?

11. A recipe for 1 person needs 3 biscuits. How many biscuits for 5 people?

12. Scale up the ratio 5:2 by multiplying both parts by 3. What is the new ratio? A) 8:5 B) 15:6 C) 15:2 D) 10:6

13. A pattern uses 2 blue beads for every 1 red bead. If there are 6 red beads, how many blue beads are there?

14. Share 6 sweets equally between 2 people. How many does each get?

15. A recipe needs 2 eggs for every 1 cake. How many eggs are needed for 6 cakes?

Standard

20 questions

1. In a fruit bowl, apples and oranges are in the ratio 1:1. There are 10 pieces of fruit in total. How many apples are there?

2. A recipe for 1 person uses 3 grams of butter. How much butter is needed for 6 people? A) 15g B) 18g C) 21g D) 24g

3. A recipe for 4 people uses 20g of sugar. How much sugar is needed for 8 people?

4. Share 16 counters in the ratio 3:1. Select ALL statements that are TRUE: A) There are 4 parts in total B) Each part is 4 C) The shares are 12 and 4 D) The shares are equal E) The larger share is 12

5. A model car is built at a scale where 1 cm represents 15 cm in real life. If the model is 3 cm long, how long is the real car?

6. In a fruit bowl, apples and oranges are in the ratio 1:3. There are 12 pieces of fruit in total. How many apples are there?

7. A recipe for 1 person uses 4 grams of butter. How much butter is needed for 7 people? A) 24g B) 28g C) 32g D) 35g

8. A recipe for 2 people uses 7g of sugar. How much sugar is needed for 6 people?

9. Share 18 counters in the ratio 1:2. How many are in the larger share?

10. A model car is built at a scale where 1 cm represents 25 cm in real life. If the model is 3 cm long, how long is the real car?

11. In a fruit bowl, apples and oranges are in the ratio 3:1. There are 16 pieces of fruit in total. How many apples are there?

12. A recipe for 1 person uses 6 grams of butter. How much butter is needed for 3 people? A) 12g B) 15g C) 18g D) 21g

13. Share 12 counters in the ratio 1:3. Select ALL statements that are TRUE: A) There are 4 parts in total B) Each part is 3 C) The shares are 3 and 9 D) The shares are 4 and 8 E) The larger share is 9

14. A model car is built at a scale where 1 cm represents 30 cm in real life. If the model is 3 cm long, how long is the real car?

15. A recipe for 2 people uses 7 grams of butter. How much butter is needed for 6 people?

16. Share 16 counters in the ratio 1:3. Select ALL statements that are TRUE: A) There are 4 parts in total B) Each part is 4 C) The shares are 4 and 12 D) The shares are equal E) The larger share is 12

17. A recipe for 2 people needs 9g of pasta. How much pasta is needed for 6 people?

18. 4 pencils cost £12. Using the same price per pencil, how much would 7 pencils cost?

19. A tap fills a container at a steady rate. In 2 minutes it fills 8 litres. Using the same rate, how many litres will it have filled after 5 minutes?

20. Share 24 pens in the ratio 1:2. How many are in the smaller share?

Hard

5 questions

1. Share 36 pens in the ratio 4:5. How many are in the larger share?

2. A drink mixes juice and water in the ratio 1:4. If there are 25ml altogether, how much is juice and how much is water?

3. A recipe for 3 people needs 15g of pasta. How much pasta is needed for 7 people?

4. 5 pencils cost £20. Using the same price per pencil, how much would 8 pencils cost?

5. A drink mixes juice and water in the ratio 3:2. If there are 25ml altogether, select ALL statements that are TRUE: A) There are 5 parts in total B) There is 15ml of juice C) There is 10ml of water D) There is 12ml of juice E) Each part is 5ml

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